skkmigas PERTAMINA EP	**PERTAMINA EP REGIONAL 1 ZONA 4**	POG
CAUSE AND EFFECT DIAGRAM	No. Dok. : TMBP-SP001-IN-LOG-D101 Rev. : A Halaman : 1 dari 13	

NAMA PEMILIK PROYEK : **PT PERTAMINA EP**

NAMA PROYEK : **JASA PEMBANGUNAN FASILITAS PRODUKSI STASIUN PENGUMPUL (SP) TANJUNG MIRING BARAT & KARANGAN TIMUR**

LOKASI : **SP TANJUNG MIRING BARAT & KARANGAN TIMUR - FIELD LIMAU**

NO. KONTRAK : **SA02035383B**

REV.	TGL.	DESKRIPSI	DISIAPKAN	DIPERIKSA	DISETUJUI	DIPERIKSA	DISETUJUI
A	05/01/2026	Issued For Review	MHP	AS	RTN		
			PT. PROSYMPAC OIL & GAS			PT. PERTAMINA EP	

TABEL REVISI

Rev.	Tanggal	Deskripsi

COMMENT RESPONSE SHEET

No.	Page	PERUSAHAAN's Comment	Contractor's Response	Remarks

DAFTAR ISI

TABEL REVISI **2**
COMMENT RESPONSE SHEET **3**
DAFTAR ISI **4**

CAUSE & EFFECT DIAGRAM
TANJUNG MIRING BARAT & KARANGAN TIMUR

NOTE

1. WILL BE DETAILED BY CONTRACTOR/VENDOR DURING EPC STAGE
2. SET POINT WILL BE DETAIL AT EPC STAGE
3. CAUSE EFFECT DIAGRAM FOR POWER PLANT AREA

SINGKATAN

AL : ALARM
BDV : BLOWDOWN VALVE
C : CLOSED
ESD : EMERGENCY SHUTDOWN
H : HIGH
HH : HIGH HIGH
HP : HIGH PRESSURE
LCP : LOCAL CONTROL PANEL
L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
USD : UNIT SHUTDOWN
SD : SHUTDOWN

CAUSE / EFFECT

CAUSE				EFFECT																											
TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	TAG NO.	SDV-V001-001	SDV-V001-002	PCV-V001-002	LCV-V001-003	SDV-V003-001	SDV-V003-002	PCV-V003-002	LCV-V003-003		SDV-V002-001	SDV-V002-002	PCV-V002-002	SDV-V003-002	SDV-V003-002	LCV-V002-003	SDV-V004-001	SDV-V004-002	PCV-V004-002	LCV-V004-003		P-002A	P-002B	P-003A	P-003B	P-001A	P-001B	ALARM
				DESCRIPTION	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	PACKAGE PRODUCTION KRG	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	SHUTDOWN VALVE	SHUTDOWN VALVE	LEVEL CONTROL VALVE	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	PACKAGE PRODUCTION KRG	TEST PUMP TMB	TEST PUMP TMB	TEST PUMP KRG	TEST PUMP KRG	GROSS TRANSFER PUMP A	GROSS TRANSFER PUMP B	
				P&ID DWG REF.	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-103	TMBP-01-PR-PID-103	TMBP-01-PR-PID-104	TMBP-01-PR-PID-104	TMBP-01-PR-PID-107	TMBP-01-PR-PID-107	
				PRODUCTION SEPARATOR TMB & TEST SEPARATOR TMB																											
PAH-TMB-001	PRESSURE ALARM HIGH	70 PSIG	TMBP-SP001-PR-PID-D101																												X
PAH-TMB-002	PRESSURE ALARM HIGH	70 PSIG	TMBP-SP001-PR-PID-D101																												X
LALL-V001-001	LEVEL ALARM LOW LOW AT V-001	6 INCH	TMBP-SP001-PR-PID-D101			C							USD																		
LAHH-V001-001	LEVEL ALARM HIGH HIGH AT V-001	30 INCH	TMBP-SP001-PR-PID-D101		C								PSD																		
PAHH-V001-001	PRESSURE ALARM HIGH HIGH AT V-001	80 PSIG	TMBP-SP001-PR-PID-D101		C								PSD																		
PIT-V001-002	PRESSURE ALARM LOW AT V-001	20 PSIG	TMBP-SP001-PR-PID-D101																												X
	PRESSURE ALARM HIGH AT V-001	70 PSIG	TMBP-SP001-PR-PID-D101																												X
LIT-V001-003	LEVEL ALARM LOW AT V-001	12 INCH	TMBP-SP001-PR-PID-D101					C																							X
--	LEVEL ALARM HIGH AT V-001	24 INCH	TMBP-SP001-PR-PID-D101					O																							X
LALL-V003-001	LEVEL ALARM LOW LOW AT V-003	6 INCH	TMBP-SP001-PR-PID-D101							C			USD																		
LAHH-V003-001	LEVEL ALARM HIGH HIGH AT V-003	30 INCH	TMBP-SP001-PR-PID-D101						C				USD																		
PAHH-V003-001	PRESSURE ALARM HIGH HIGH AT V-003	50 PSIG	TMBP-SP001-PR-PID-D101						C				USD																		
PIT-V003-002	PRESSURE ALARM LOW AT V-003	20 PSIG	TMBP-SP001-PR-PID-D101																												X
--	PRESSURE ALARM HIGH AT V-003	70 PSIG	TMBP-SP001-PR-PID-D101																												X
LIT-V003-003	LEVEL ALARM LOW AT V-003	12 INCH	TMBP-SP001-PR-PID-D101																												X
--	LEVEL ALARM HIGH AT V-003	24 INCH	TMBP-SP001-PR-PID-D101																												X
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D101		C	C	O		C	C	O																				
--				**PRODUCTION SEPARATOR KRG & TEST SEPARATOR KRG**																											
PAH-KRG-001	PRESSURE ALARM HIGH	70 PSIG	TMBP-SP001-PR-PID-D101																												X
PAH-KRG-002	PRESSURE ALARM HIGH	70 PSIG	TMBP-SP001-PR-PID-D101																												X
LALL-V002-001	LEVEL ALARM LOW LOW AT V-002	6 INCH	TMBP-SP001-PR-PID-D102												C									USD							
LAHH-V002-001	LEVEL ALARM HIGH HIGH AT V-002	30 INCH	TMBP-SP001-PR-PID-D102											C										PSD							
PAHH-V002-001	PRESSURE ALARM HIGH HIGH AT V-002	80 PSIG	TMBP-SP001-PR-PID-D102											C										PSD							
PIT-V002-002	PRESSURE ALARM LOW AT V-002	20 PSIG	TMBP-SP001-PR-PID-D102																												
--	PRESSURE ALARM HIGH AT V-002	70 PSIG	TMBP-SP001-PR-PID-D102																												
LIT-V002-003	LEVEL ALARM LOW AT V-002	12 INCH	TMBP-SP001-PR-PID-D102																												
--	LEVEL ALARM HIGH AT V-002	24 INCH	TMBP-SP001-PR-PID-D102																												
LALL-V004-001	LEVEL ALARM LOW LOW AT V-004	6 INCH	TMBP-SP001-PR-PID-D102															C			C			USD							
LAHH-V004-001	LEVEL ALARM HIGH HIGH AT V-004	30 INCH	TMBP-SP001-PR-PID-D102														C			C				USD							
PAHH-V004-001	PRESSURE ALARM HIGH HIGH AT V-004	50 PSIG	TMBP-SP001-PR-PID-D102														C			C				USD							

CAUSE & EFFECT DIAGRAM
TANJUNG MIRING BARAT & KARANGAN TIMUR

NOTE

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H : HIGH
HH : HIGH HIGH
HP : HIGH PRESSURE
LCP : LOCAL CONTROL PANEL
L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
USD : UNIT SHUTDOWN
SD : SHUTDOWN

CAUSE

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.
PIT-V004-002	PRESSURE ALARM LOW AT V-004	20 PSIG	TMBP-SP001-PR-PID-D102
--	PRESSURE ALARM HIGH AT V-004	70 PSIG	TMBP-SP001-PR-PID-D102
LIT-V004-003	LEVEL ALARM LOW AT V-004	12 INCH	TMBP-SP001-PR-PID-D102
--	LEVEL ALARM HIGH AT V-004	24 INCH	TMBP-SP001-PR-PID-D102
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D102
--			
LALL-T003-001	LEVEL ALARM LOW LOW AT T-005A	1.8 FT	TMBP-SP001-PR-PID-D103
LAHH-T004-001	LEVEL ALARM HIGH HIGH AT T-005B	1.81 FT	TMBP-SP001-PR-PID-D104
LIT-T003-002	PRESSURE ALARM LOW AT V-005A	2.41 FT	TMBP-SP001-PR-PID-D103
--	PRESSURE ALARM HIGH AT V-005A	10.02 FT	TMBP-SP001-PR-PID-D103
LIT-T004-003	LEVEL ALARM LOW AT V-005B	2.31 FT	TMBP-SP001-PR-PID-D104
--	LEVEL ALARM HIGH AT V-005B	10.07 FT	TMBP-SP001-PR-PID-D104
PALL-P002-001A	PRESSURE ALARM LOW LOW AT P-002A	-1 PSIG	TMBP-SP001-PR-PID-D103
PAHH-P002-002A	PRESSURE ALARM HIGH HIGH AT P-002A	45 PSIG	TMBP-SP001-PR-PID-D103
PALL-P002-001B	PRESSURE ALARM LOW LOW AT P-002B	-1 PSIG	TMBP-SP001-PR-PID-D103
PAHH-P002-002B	PRESSURE ALARM HIGH HIGH AT P-002B	45 PSIG	TMBP-SP001-PR-PID-D103
PALL-P003-001A	PRESSURE ALARM LOW LOW AT P-003A	-1 PSIG	TMBP-SP001-PR-PID-D104
PAHH-P003-002A	PRESSURE ALARM HIGH HIGH AT P-003A	45 PSIG	TMBP-SP001-PR-PID-D104
PALL-P003-001B	PRESSURE ALARM LOW LOW AT P-003B	-1 PSIG	TMBP-SP001-PR-PID-D104
PAHH-P003-002B	PRESSURE ALARM HIGH HIGH AT P-003B	45 PSIG	TMBP-SP001-PR-PID-D104
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D103&104
LALL-T005-001A	LEVEL ALARM LOW LOW AT T-005-001A	2.3 FT	TMBP-SP001-PR-PID-D105
LALL-T005-001B	LEVEL ALARM LOW LOW AT T-005-001B	4.5 FT	TMBP-SP001-PR-PID-D105
LIT-T005-001A	PRESSURE ALARM LOW AT V-005A-001A	3.4 FT	TMBP-SP001-PR-PID-D105
--	PRESSURE ALARM HIGH AT V-005A-001A	12.33 FT	TMBP-SP001-PR-PID-D105
LIT-T005-001B	LEVEL ALARM LOW AT V-005A-001B	5.6 FT	TMBP-SP001-PR-PID-D105
--	LEVEL ALARM HIGH AT V-005A-001B	14.53 FT	TMBP-SP001-PR-PID-D106
LALL-T006-001A	LEVEL ALARM LOW LOW AT T-006A	1.98 FT	TMBP-SP001-PR-PID-D106
LALL-T006-001B	LEVEL ALARM LOW LOW AT T-006B	1.98 FT	TMBP-SP001-PR-PID-D106
LIT-T006-001A	PRESSURE ALARM LOW AT V-006A-001A	2.48 FT	TMBP-SP001-PR-PID-D106

EFFECT

TAG NO.	DESCRIPTION	P&ID DWG REF.																												
		TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-101	TMBP-01-PR-PID-101	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-102	TMBP-01-PR-PID-103	TMBP-01-PR-PID-103	TMBP-01-PR-PID-104	TMBP-01-PR-PID-104	TMBP-01-PR-PID-107	TMBP-01-PR-PID-107	
	DESCRIPTION	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	PACKAGE PRODUCTION KRG	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	SHUTDOWN VALVE	SHUTDOWN VALVE	LEVEL CONTROL VALVE	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	PACKAGE PRODUCTION KRG	TEST PUMP TMB	TEST PUMP TMB	TEST PUMP KRG	TEST PUMP KRG	GROSS TRANSFER PUMP A	GROSS TRANSFER PUMP B	
	TAG NO.	SDV-V001-001	SDV-V001-002	PCV-V001-002	LCV-V001-003	SDV-V003-001	SDV-V003-002	PCV-V003-002	LCV-V003-003		SDV-V002-001	SDV-V002-002	PCV-V002-002	SDV-V003-002	SDV-V003-002	LCV-V002-003	SDV-V004-001	SDV-V004-002	PCV-V004-002	LCV-V004-003		P-002A	P-002B	P-003A	P-003B	P-001A	P-001B	ALARM
PIT-V004-002	PRESSURE ALARM LOW AT V-004																												
--	PRESSURE ALARM HIGH AT V-004																												
LIT-V004-003	LEVEL ALARM LOW AT V-004																												
--	LEVEL ALARM HIGH AT V-004																												
ESD--	EMERGENCY PUSH BUTTON INTERLOCK										C	C	O	C	C		C	C	O										
--		**TEST TANK TMB & KRG**																											
LALL-T003-001	LEVEL ALARM LOW LOW AT T-005A														C							STO	STO						
LAHH-T004-001	LEVEL ALARM HIGH HIGH AT T-005B																	C						STO	STO				
LIT-T003-002	PRESSURE ALARM LOW AT V-005A																												X
--	PRESSURE ALARM HIGH AT V-005A																												X
LIT-T004-003	LEVEL ALARM LOW AT V-005B																												X
--	LEVEL ALARM HIGH AT V-005B																												X
PALL-P002-001A	PRESSURE ALARM LOW LOW AT P-002A																					STO							
PAHH-P002-002A	PRESSURE ALARM HIGH HIGH AT P-002A																					STO							
PALL-P002-001B	PRESSURE ALARM LOW LOW AT P-002B																						STO						
PAHH-P002-002B	PRESSURE ALARM HIGH HIGH AT P-002B																						STO						
PALL-P003-001A	PRESSURE ALARM LOW LOW AT P-003A																							STO					
PAHH-P003-002A	PRESSURE ALARM HIGH HIGH AT P-003A																							STO					
PALL-P003-001B	PRESSURE ALARM LOW LOW AT P-003B																								STO				
PAHH-P003-002B	PRESSURE ALARM HIGH HIGH AT P-003B																								STO				
ESD--	EMERGENCY PUSH BUTTON INTERLOCK																					SD	SD	SD	SD				
		WASH TANK TMB & GROSS TRANSFER PUMP																											
LALL-T005-001A	LEVEL ALARM LOW LOW AT T-005-001A																									STO	STO		
LALL-T005-001B	LEVEL ALARM LOW LOW AT T-005-001B																									STO	STO		
LIT-T005-001A	PRESSURE ALARM LOW AT V-005A-001A																												X
--	PRESSURE ALARM HIGH AT V-005A-001A																												X
LIT-T005-001B	LEVEL ALARM LOW AT V-005A-001B																												X
--	LEVEL ALARM HIGH AT V-005A-001B																												X
LALL-T006-001A	LEVEL ALARM LOW LOW AT T-006A																									STO	STO		
LALL-T006-001B	LEVEL ALARM LOW LOW AT T-006B																									STO	STO		
LIT-T006-001A	PRESSURE ALARM LOW AT V-006A-001A																												X

CAUSE & EFFECT DIAGRAM
TANJUNG MIRING BARAT & KARANGAN TIMUR

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L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
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SD : SHUTDOWN

EFFECT

TAG NO.	DESCRIPTION	P&ID DWG REF.
SDV-V001-001	SHUTDOWN VALVE	TMBP-01-PR-PID-101
SDV-V001-002	SHUTDOWN VALVE	TMBP-01-PR-PID-101
PCV-V001-002	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-101
LCV-V001-003	LEVEL CONTROL VALVE	TMBP-01-PR-PID-101
SDV-V003-001	SHUTDOWN VALVE	TMBP-01-PR-PID-101
SDV-V003-002	SHUTDOWN VALVE	TMBP-01-PR-PID-101
PCV-V003-002	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-101
LCV-V003-003	LEVEL CONTROL VALVE	TMBP-01-PR-PID-101
	PACKAGE PRODUCTION KRG	TMBP-01-PR-PID-101
SDV-V002-001	SHUTDOWN VALVE	TMBP-01-PR-PID-102
SDV-V002-002	SHUTDOWN VALVE	TMBP-01-PR-PID-102
PCV-V002-002	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-102
SDV-V003-002	SHUTDOWN VALVE	TMBP-01-PR-PID-101
SDV-V003-002	SHUTDOWN VALVE	TMBP-01-PR-PID-101
LCV-V002-003	LEVEL CONTROL VALVE	TMBP-01-PR-PID-102
SDV-V004-001	SHUTDOWN VALVE	TMBP-01-PR-PID-102
SDV-V004-002	SHUTDOWN VALVE	TMBP-01-PR-PID-102
PCV-V004-002	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-102
LCV-V004-003	LEVEL CONTROL VALVE	TMBP-01-PR-PID-102
	PACKAGE PRODUCTION KRG	TMBP-01-PR-PID-102
P-002A	TEST PUMP TMB	TMBP-01-PR-PID-103
P-002B	TEST PUMP TMB	TMBP-01-PR-PID-103
P-003A	TEST PUMP KRG	TMBP-01-PR-PID-104
P-003B	TEST PUMP KRG	TMBP-01-PR-PID-104
P-001A	GROSS TRANSFER PUMP A	TMBP-01-PR-PID-107
P-001B	GROSS TRANSFER PUMP B	TMBP-01-PR-PID-107
ALARM		

CAUSE

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	ALARM
--	PRESSURE ALARM HIGH AT V-006A-001A	12.14 FT	TMBP-SP001-PR-PID-D106	X
LIT-T006-001B	LEVEL ALARM LOW AT V-006A-001B	2.48 FT	TMBP-SP001-PR-PID-D106	X
--	LEVEL ALARM HIGH AT V-006A-001B	12.14 FT	TMBP-SP001-PR-PID-D106	X
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D105&106	

TAG NO.	P-001A	P-001B
ESD--	SD	SD

CAUSE & EFFECT DIAGRAM
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O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
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EFFECT

TAG NO.	DESCRIPTION	P&ID DWG REF.
P-004A	FEED WATER PUMP	TMBP-01-PR-PID-108
P-004B	FEED WATER PUMP	TMBP-01-PR-PID-108
P-009A/B	PH ADJUSTMENT DOSING PUMP	TMBP-01-PR-PID-111
P-010A/B	REVERSE DEMULSIFIER DOSING PUMP	TMBP-01-PR-PID-111
P-011A/B	FOLCULANT DOSING PUMP	TMBP-01-PR-PID-111
P-012A/B	BIOCIDE DOSING PUMP	TMBP-01-PR-PID-111
P-013A/B	CORRISION INHIBITOR DOSING PUMP	TMBP-01-PR-PID-111
P-014A/B	SCALE INHIBITOR DOSING PUMP	TMBP-01-PR-PID-111
P-015A/B	OXYGEN SCAVENGER DOSING PUMP	TMBP-01-PR-PID-111
	ALL CHEMICAL PACKAGE 1	
P-005A	INJECTION PUMP A	TMBP-01-PR-PID-113
P-005B	INJECTION PUMP B	TMBP-01-PR-PID-113
SDV-P005	SHUTDOWN VALVE	TMBP-01-PR-PID-113
FCV-P005	FLOW CONTROL VALVE	TMBP-01-PR-PID-113
PCV-V005	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-114
SDV-V005	SHUTDOWN VALVE	TMBP-01-PR-PID-114
LCV-V005	LEVEL CONTROL VALVE	TMBP-01-PR-PID-114
SDV-V005	SHUTDOWN VALVE	TMBP-01-PR-PID-114
BDV-V005	BLOW DOWN VALVE	TMBP-01-PR-PID-114
P-006A	KNOCK OUT DRUM PUMP A	TMBP-01-PR-PID-115
P-006B	KNOCK OUT DRUM PUMP B	TMBP-01-PR-PID-115
ALARM		

CAUSE

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	P-004A	P-004B	P-009A/B	P-010A/B	P-011A/B	P-012A/B	P-013A/B	P-014A/B	P-015A/B	ALL CHEMICAL PACKAGE 1	P-005A	P-005B	SDV-P005	FCV-P005	PCV-V005	SDV-V005	LCV-V005	SDV-V005	BDV-V005	P-006A	P-006B	ALARM
													CHEMICAL PACKAGE 1												
LALL-T009-001	LEVEL ALARM LOW LOW AT T-009	Note 2	TMBP-SP001-PR-PID-D111			STO																			
PAHH-T009-001	PRESSURE ALARM HIGH HIGH AT T-009	Note 2	TMBP-SP001-PR-PID-D111			STO																			
LALL-T010-001	LEVEL ALARM LOW LOW AT T-010	Note 2	TMBP-SP001-PR-PID-D111				STO																		
PAHH-T010-001	PRESSURE ALARM HIGH HIGH AT T-010	Note 2	TMBP-SP001-PR-PID-D111				STO																		
LALL-T011-001	LEVEL ALARM LOW LOW AT T-011	Note 2	TMBP-SP001-PR-PID-D111					STO																	
PAHH-T011-001	PRESSURE ALARM HIGH HIGH AT T-011	Note 2	TMBP-SP001-PR-PID-D111					STO																	
LALL-T012-001	LEVEL ALARM LOW LOW AT T-012	Note 2	TMBP-SP001-PR-PID-D111						STO																
PAHH-T012-001	PRESSURE ALARM HIGH HIGH AT T-012	Note 2	TMBP-SP001-PR-PID-D111						STO																
LALL-T013-001	LEVEL ALARM LOW LOW AT T-013	Note 2	TMBP-SP001-PR-PID-D111							STO															
PAHH-T013-001	PRESSURE ALARM HIGH HIGH AT T-013	Note 2	TMBP-SP001-PR-PID-D111							STO															
LALL-T014-001	LEVEL ALARM LOW LOW AT T-014	Note 2	TMBP-SP001-PR-PID-D111								STO														
PAHH-T014-001	PRESSURE ALARM HIGH HIGH AT T-014	Note 2	TMBP-SP001-PR-PID-D111								STO														
LALL-T015-001	LEVEL ALARM LOW LOW AT T-015	Note 2	TMBP-SP001-PR-PID-D111									STO													
PAHH-T015-001	PRESSURE ALARM HIGH HIGH AT T-015	Note 2	TMBP-SP001-PR-PID-D111									STO													
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D111										STO												
													PRODUCTION SCRUBBER												
PIT-V005-001	PRESSURE ALARM LOW AT V-005	Note 2	TMBP-SP001-PR-PID-D114														C								
--	PRESSURE ALARM HIGH AT V-005	Note 2	TMBP-SP001-PR-PID-D114														O								
LALL-V005-001	LEVEL ALARM LOW LOW V-008	Note 2	TMBP-SP001-PR-PID-D114															C							
LAL-V005-002	LEVEL ALARM LOW AT V-005	Note 2	TMBP-SP001-PR-PID-D114																C						
LAH-V005-002	LEVEL ALARM HIGH AT V-005	Note 2	TMBP-SP001-PR-PID-D114																O						
ESD-V005-002	EMERGENCY PUSH BUTTON		TMBP-SP001-PR-PID-D114																	C					
ESD-V005-001	EMERGENCY PUSH BUTTON		TMBP-SP001-PR-PID-D114															C			O				

CAUSE & EFFECT DIAGRAM
TANJUNG MIRING BARAT & KARANGAN TIMUR

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SINGKATAN

AL : ALARM
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C : CLOSED
ESD : EMERGENCY SHUTDOWN
H : HIGH
HH : HIGH HIGH
HP : HIGH PRESSURE
LCP : LOCAL CONTROL PANEL
L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
USD : UNIT SHUTDOWN
SD : SHUTDOWN

CAUSE

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.
LAL-V006-001	LEVEL ALARM LOW AT V-006	Note 2	TMBP-SP001-PR-PID-D115
LAH-V006-001	LEVEL ALARM HIGH AT V-006	Note 2	TMBP-SP001-PR-PID-D115
PALL-V006-001A	PRESSURE ALARM LOW LOW AT P-006A	-1 PSIG	TMBP-SP001-PR-PID-D115
PAHH-V006-002A	PRESSURE ALARM HIGH HIGH AT P-006A	35 PSIG	TMBP-SP001-PR-PID-D115
PALL-V006-001B	PRESSURE ALARM LOW LOW AT P-006B	-1 PSIG	TMBP-SP001-PR-PID-D115
PAHH-V006-002B	PRESSURE ALARM HIGH HIGH AT P-006B	35 PSIG	TMBP-SP001-PR-PID-D115
LALL-V005-001	LEVEL ALARM LOW LOW V-006	Note 2	TMBP-SP001-PR-PID-D115
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D115

EFFECT

TAG NO.	DESCRIPTION	P&ID DWG REF.	KNOCK OUT DRUM																					
P-004A	FEED WATER PUMP	TMBP-01-PR-PID-108																				STO	STO	
P-004B	FEED WATER PUMP	TMBP-01-PR-PID-108																				STA	STA	
P-009A/B	PH ADJUSTMENT DOSING PUMP	TMBP-01-PR-PID-111																				STO		
P-010A/B	REVERSE DEMULSIFIER DOSING PUMP	TMBP-01-PR-PID-111																				STO		
P-011A/B	FOLCULANT DOSING PUMP	TMBP-01-PR-PID-111																					STO	
P-012A/B	BIOCIDE DOSING PUMP	TMBP-01-PR-PID-111																					STO	
P-013A/B	CORRISION INHIBITOR DOSING PUMP	TMBP-01-PR-PID-111																						
P-014A/B	SCALE INHIBITOR DOSING PUMP	TMBP-01-PR-PID-111																				STO	STO	
P-015A/B	OXYGEN SCAVENGER DOSING PUMP	TMBP-01-PR-PID-111																				STO	STO	
	ALL CHEMICAL PACKAGE 1																							
P-005A	INJECTION PUMP A	TMBP-01-PR-PID-113																						
P-005B	INJECTION PUMP B	TMBP-01-PR-PID-113																						
SDV-P005	SHUTDOWN VALVE	TMBP-01-PR-PID-113																						
FCV-P005	FLOW CONTROL VALVE	TMBP-01-PR-PID-113																						
PCV-V005	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-114																						
SDV-V005	SHUTDOWN VALVE	TMBP-01-PR-PID-114																						
LCV-V005	LEVEL CONTROL VALVE	TMBP-01-PR-PID-114																						
SDV-V005	SHUTDOWN VALVE	TMBP-01-PR-PID-114																						
BDV-V005	BLOW DOWN VALVE	TMBP-01-PR-PID-114																						
P-006A	KNOCK OUT DRUM PUMP A	TMBP-01-PR-PID-115																				STO	STO	
P-006B	KNOCK OUT DRUM PUMP B	TMBP-01-PR-PID-115																				SD	SD	
ALARM																								

CAUSE & EFFECT DIAGRAM
TANJUNG MIRING BARAT & KARANGAN TIMUR

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HH : HIGH HIGH
HP : HIGH PRESSURE
LCP : LOCAL CONTROL PANEL
L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
USD : UNIT SHUTDOWN
SD : SHUTDOWN

CAUSE / EFFECT

CAUSE				EFFECT																						
TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	TAG NO.	DESCRIPTION	P&ID DWG REF.																				
				P-007A	CLOSED DRAIN PUMP A	TMBP-01-PR-PID-116																				
				P-007B	CLOSED DRAIN PUMP B	TMBP-01-PR-PID-116																				
				P-020	SLURRY PUMP	TMBP-01-PR-PID-117																				
				P-019	WATER SUMP PUMP	TMBP-01-PR-PID-117																				
				P-022	OIL SUMP PUMP	TMBP-01-PR-PID-117																				
				SDV-V008-001	SHUTDOWN VALVE	TMBP-01-PR-PID-118																				
				SDV-V008-002	SHUTDOWN VALVE	TMBP-01-PR-PID-118																				
				PCV-V008-002	PRESSURE CONTROL VALVE	TMBP-01-PR-PID-118																				
				LCV-V008-003	LEVEL CONTROL VALVE	TMBP-01-PR-PID-118																				
				SDV-Z002-002	SHUTDOWN VALVE	TMBP-01-PR-PID-118																				
				BDV-Z002-001	BLOW DOWN VALVE	TMBP-01-PR-PID-118																				
					FUEL GAS SCRUBBER PACKAGE	TMBP-01-PR-PID-118																				
					ALL FUEL GAS SYSTEM	TMBP-01-PR-PID-118																				
				P-016A	BOOSTER PUMP A	TMBP-01-PR-PID-121																				
				P-016B	BOOSTER PUMP B	TMBP-01-PR-PID-121																				
				P-017A	WATER DISTRIBUTION PUMP A	TMBP-01-PR-PID-121																				
				P-017B	WATER DISTRIBUTION PUMP B	TMBP-01-PR-PID-121																				
				P-008A	DEEP WELL PUMP A	TMBP-01-PR-PID-121																				
				P-008B	DEEP WELL PUMP B	TMBP-01-PR-PID-121																				
					ALL AIR COMPRESSOR PACKAGE																					
					AIR COMPRESSOR																					
					ALARM																					
					CLOSED DRAIN VESSEL																					
LAL-V007-001	LEVEL ALARM LOW AT V-007	12 INCH	TMBP-SP001-PR-PID-D116			STO	STO																			
LAH-V007-001	LEVEL ALARM HIGH AT V-007	48 INCH	TMBP-SP001-PR-PID-D116			STA	STA																			
LALL-V007-001	LEVEL ALARM LOW LOW V-007	6 INCH	TMBP-SP001-PR-PID-D116			STO	STO																			
					SUMP PIT & PIT PUMP																					
LAL-PN001-002	LEVEL ALARM LOW AT PN-001	Note 2	TMBP-SP001-PR-PID-D117						STO																	
LAH-PN001-002	LEVEL ALARM HIGH AT PN-001	Note 2	TMBP-SP001-PR-PID-D117						STA																	
					FUEL GAS SYSTEM																					
PAHH-V008-001	PRESSURE ALARM HIGH HIGH AT V-008	Note 2	TMBP-SP001-PR-PID-D118								C															
LAHH-V008-002	LEVEL ALARM HIGH HIGH AT V-008	Note 2	TMBP-SP001-PR-PID-D118								C						USD									
LALL-V008-002	LEVEL ALARM LOW LOW V-008	Note 2	TMBP-SP001-PR-PID-D118									C					USD									
PIT-V008-002	PRESSURE ALARM LOW AT V-008	Note 2	TMBP-SP001-PR-PID-D118																							
--	PRESSURE ALARM HIGH AT V-008	Note 2	TMBP-SP001-PR-PID-D118																							
LIT-V008-003	LEVEL ALARM LOW AT V-008	Note 2	TMBP-SP001-PR-PID-D118																							
--	LEVEL ALARM HIGH AT V-008	Note 2	TMBP-SP001-PR-PID-D118																							
ESD-Z002-001	EMERGENCY PUSH BUTTON		TMBP-SP001-PR-PID-D118								C	C				O										
ESD-Z002-002	EMERGENCY PUSH BUTTON		TMBP-SP001-PR-PID-D118												C											
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D118															SD								
					UTILITY WATER SYSTEM																					
LAL-T017-002	LEVEL ALARM LOW AT T-017	Note 2	TMBP-SP001-PR-PID-D121																STA	STA	STO	STO				
LAH-T017-001	LEVEL ALARM HIGH AT T-017	Note 2	TMBP-SP001-PR-PID-D121																STO	STO						
PAL-T017-001	PRESSURE ALARM LOW AT T-017	Note 2	TMBP-SP001-PR-PID-D121																		STA	STA				
PAH-T017-001	PRESSURE ALARM HIGH AT T-017	Note 2	TMBP-SP001-PR-PID-D121																		STO	STO				
					UTILITY AIR AND INSTRUMET AIR SYSTEM																					
ESD--	EMERGENCY PUSH BUTTON INTERLOCK		TMBP-SP001-PR-PID-D120																					SD	SD	

CAUSE & EFFECT DIAGRAM
TANJUNG MIRING BARAT & KARANGAN TIMUR

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HH : HIGH HIGH
HP : HIGH PRESSURE
LCP : LOCAL CONTROL PANEL
L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
USD : UNIT SHUTDOWN
SD : SHUTDOWN

CAUSE / EFFECT

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	P&ID DWG REF.	DESCRIPTION	TAG NO.																			
				TMBP-01-PR-PID-116	CLOSED DRAIN PUMP A	P-007A	TMBP-01-PR-PID-116	TMBP-01-PR-PID-117	TMBP-01-PR-PID-117	TMBP-01-PR-PID-117	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-118	TMBP-01-PR-PID-121	TMBP-01-PR-PID-121	TMBP-01-PR-PID-121	TMBP-01-PR-PID-121	TMBP-01-PR-PID-121	TMBP-01-PR-PID-121			
							CLOSED DRAIN PUMP B	SLURRY PUMP	WATER SUMP PUMP	OIL SUMP PUMP	SHUTDOWN VALVE	SHUTDOWN VALVE	PRESSURE CONTROL VALVE	LEVEL CONTROL VALVE	SHUTDOWN VALVE	BLOW DOWN VALVE	FUEL GAS SCRUBBER PACKAGE	ALL FUEL GAS SYSTEM	BOOSTER PUMP A	BOOSTER PUMP B	WATER DISTRIBUTION PUMP A	WATER DISTRIBUTION PUMP B	DEEP WELL PUMP A	DEEP WELL PUMP B	ALL AIR COMPRESSOR PACKAGE	AIR COMPRESSOR	ALARM
							P-007B	P-020	P-019	P-022	SDV-V008-001	SDV-V008-002	PCV-V008-002	LCV-V008-003	SDV-Z002-002	BDV-Z002-001			P-016A	P-016B	P-017A	P-017B	P-008A	P-008B			

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HP : HIGH PRESSURE
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L : LOW
LL : LOW LOW
LP : LOW PRESSURE
O : OPEN
PACK : PACKAGE
PSD : PROCESS SHUTDOWN
STA : START
STO : STOP
SDV : SHUTDOWN VALVE
USD : UNIT SHUTDOWN
SD : SHUTDOWN

EFFECT

TAG NO.	DESCRIPTION	P&ID DWG REF.
DV-0001	DELUGE VALVE-001	TMBP-01-PR-PID-116
DV-0002	DELUGE VALVE-002	TMBP-01-PR-PID-116
DV-0003	DELUGE VALVE-003	TMBP-01-PR-PID-117
DV-0004	DELUGE VALVE-004	TMBP-01-PR-PID-117
DV-0005	DELUGE VALVE-005	TMBP-01-PR-PID-117
DV-0006	DELUGE VALVE-006	TMBP-01-PR-PID-118
DV-0007	DELUGE VALVE-007	TMBP-01-PR-PID-118
DV-0008	DELUGE VALVE-008	TMBP-01-PR-PID-118
DV-0009	DELUGE VALVE-009	TMBP-01-PR-PID-118
DV-0010	DELUGE VALVE-010	TMBP-01-PR-PID-118
DV-0011	DELUGE VALVE-011	TMBP-01-PR-PID-118
	FOAM	TMBP-01-PR-PID-118
	FM-200	TMBP-01-PR-PID-118
	FIRE SUSPRESSION	TMBP-01-PR-PID-118
	ALARM	TMBP-01-PR-PID-118

CAUSE

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	FIRE AND GAS														
GD-017	GAS DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
FD-001	FLAME DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
FD-002	FLAME DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
FD-003	FLAME DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
FD-004	FLAME DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
FD-005	FLAME DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
FD-006	FLAME DETECTOR ALARM		TMBP-SP001-IN-LAY-D108															X
LHD-001	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108		STA										STA			X
LHD-002	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108	STA											STA			X
LHD-003	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108			STA									STA			X
LHD-004	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108				STA								STA			X
LHD-005	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108											STA	STA			X
LHD-006	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108								STA				STA			X
LHD-007	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108							STA					STA			X
LHD-008	LINE HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D108						STA						STA			X
SD-018	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D108														STA	X
SD-019	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D108														STA	X
SD-020	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D108														STA	X
SD-001	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109													STA		X
SD-002	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109															X
SD-003	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109													STA		X
SD-004	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109													STA		X
SD-005	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109													STA		X

CAUSE & EFFECT DIAGRAM
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CAUSE / EFFECT

TAG NO.	DESCRIPTION	SET POINT	P&ID NO.	TAG NO.	DESCRIPTION	P&ID DWG REF.	FIRE AND GAS														
SD-006	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0001	DELUGE VALVE-001	TMBP-01-PR-PID-116											STA		X
SD-007	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0002	DELUGE VALVE-002	TMBP-01-PR-PID-116													X
SD-008	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0003	DELUGE VALVE-003	TMBP-01-PR-PID-117													X
SD-009	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0004	DELUGE VALVE-004	TMBP-01-PR-PID-117													X
SD-010	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0005	DELUGE VALVE-005	TMBP-01-PR-PID-117													X
SD-011	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0006	DELUGE VALVE-006	TMBP-01-PR-PID-118													X
SD-012	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0007	DELUGE VALVE-007	TMBP-01-PR-PID-118													X
SD-013	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0008	DELUGE VALVE-008	TMBP-01-PR-PID-118													X
SD-014	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0009	DELUGE VALVE-009	TMBP-01-PR-PID-118													X
SD-015	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0010	DELUGE VALVE-010	TMBP-01-PR-PID-118													X
SD-016	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109	DV-0011	DELUGE VALVE-011	TMBP-01-PR-PID-118													X
SD-017	SMOKE DETECTOR ALARM		TMBP-SP001-IN-LAY-D109		FOAM	TMBP-01-PR-PID-118													X
					FM-200	TMBP-01-PR-PID-118													
HD-001	HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D109		FIRE SUSPRESSION	TMBP-01-PR-PID-118													X
HD-002	HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D109		ALARM	TMBP-01-PR-PID-118													X
HD-003	HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D109																X
HD-004	HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D109																X
HD-005	HEAT DETECTOR ALARM		TMBP-SP001-IN-LAY-D109																X
MAC-001	MANUAL CALL POINT ALARM		TMBP-SP001-IN-LAY-D109																X
MAC-002	MANUAL CALL POINT ALARM		TMBP-SP001-IN-LAY-D109																X
MAC-003	MANUAL CALL POINT ALARM		TMBP-SP001-IN-LAY-D109																X
MAC-004	MANUAL CALL POINT ALARM		TMBP-SP001-IN-LAY-D109																X
MAC-005	MANUAL CALL POINT ALARM		TMBP-SP001-IN-LAY-D109																X